

Unit 32: Exponential Functions

1. The idea, in plain words

Meet the single most well-behaved function in all of calculus: e^x . Its derivative is itself. Not a scaled version, not a shifted version — literally, exactly the same function you started with:

$$\frac{d}{dx}[e^x] = e^x$$

Picture what this means: at every single point on the graph of $y = e^x$, the height of the curve and the slope of the curve are the same number. If the curve is at height 5, its slope right there is also 5. This is genuinely unique — no other basic function behaves this way — and it's exactly why e (a special constant, ≈ 2.71828) shows up everywhere in calculus, from compound interest to population growth to radioactive decay.

A warning about a very natural, very wrong instinct: because you're so used to the power rule bringing an exponent down front, it's tempting to think $\frac{d}{dx}[e^x]$ should involve “bringing down” something. It doesn't. e^x isn't x raised to a power — it's a constant raised to a variable power, which is a completely different kind of function, and it plays by its own (much simpler) rule.

Since e^x is a function like any other, the chain rule still applies whenever the exponent is something more complicated than plain x — you just multiply by the derivative of that exponent, exactly like every other composed function you've handled.

Because $\ln x$ and e^x are inverse functions of each other (from Unit 31), solving an equation with e raised to some expression is just a matter of taking $\ln$ of both sides to pull the exponent back down to ground level.

2. Toolbox

Derivative:

$$\frac{d}{dx}[e^x] = e^x$$

Chain rule version (for e raised to anything more complex than plain x):

$$\frac{d}{dx}[e^u] = e^u \cdot u'$$

Antiderivative:

$$\int e^x dx = e^x + C \qquad \int e^u \cdot u' dx = e^u + C \text{ (via substitution)}$$

Basic exponent properties (same rules as any exponent):

$$e^0 = 1 \qquad e^a \cdot e^b = e^{a+b} \qquad \frac{e^a}{e^b} = e^{a-b} \qquad (e^a)^b = e^{ab}$$

Inverse relationship with $\ln$:

$$e^{\ln x} = x \qquad \ln(e^x) = x$$

Solving an equation with e raised to an expression: isolate the exponential term first if needed, then take $\ln$ of both sides to bring the exponent down:

$$e^{(\text{expression})} = k \implies \text{expression} = \ln k$$

3. Common mistakes

- Trying to apply the power rule to e^x . $\frac{d}{dx}[e^x]$ is just e^x — there's no exponent to “bring down” and no reduction in power. This is the single most common error in this entire unit.
- Forgetting the chain rule multiplier when the exponent isn't plain x . $\frac{d}{dx}[e^{3x}]$ is $3e^{3x}$, not just e^{3x} — you must multiply by the derivative of the exponent.
- Forgetting to isolate the exponential term before taking $\ln$. If there's addition or subtraction happening around the $e^{(\cdot)}$ term (like $2e^x - 3 = 7$), you must isolate the exponential piece by itself before taking the log of anything.
- Sign errors in the exponent when solving equations. Take care carrying negative signs through, especially in decay problems.
- Forgetting that $\int e^x dx = e^x + C$ doesn't follow the power-rule pattern. There's no “+1” to the exponent and no dividing by a new exponent — e^x is simply its own antiderivative too.

4. Problem Set

Warm-up

1. $\frac{d}{dx}[e^x]$
2. $\frac{d}{dx}[e^{2x}]$
3. $\frac{d}{dx}[3e^x]$
4. Solve for x : $e^x = 5$.
5. $\int e^x dx$
6. $\frac{d}{dx}[e^x + x^2]$

Standard

7. $\frac{d}{dx}[e^{x^2}]$
8. $\frac{d}{dx}[e^{3x+1}]$
9. $\frac{d}{dx}[xe^x]$
10. $\frac{d}{dx}\left[\frac{e^x}{x}\right]$
11. Solve for x : $e^{2x-1} = 7$.
12. $\int 3e^{2x} dx$
13. $\int_0^1 e^x dx$

Challenge

14. Solve for x : $e^{3x-6} = 9$.
15. $y = (9x^2 - 6x + 2)e^{3x}$. Find $\frac{dy}{dx}$.
16. $\frac{d}{dx}[e^{\sin x}]$
17. $\int 2xe^{x^2} dx$
18. Solve for x : $2e^x - 3 = 7$.

Applied

19. An investment grows according to $A(t) = 1000e^{0.06t}$ (continuously compounded). Find the time t when the investment doubles.
20. A population grows according to $P(t) = 200e^{0.03t}$. Find the growth rate function $\frac{dP}{dt}$, and evaluate it at $t = 10$.
21. A drug's concentration in the bloodstream is $C(t) = 50e^{-0.5t}$ mg. Find the rate of change of concentration, $\frac{dC}{dt}$, at $t = 2$.
22. A cooling object's temperature difference from room temperature is modeled by $D(t) = 80e^{-0.1t}$. Find the total accumulated area under this curve from $t = 0$ to $t = 10$.